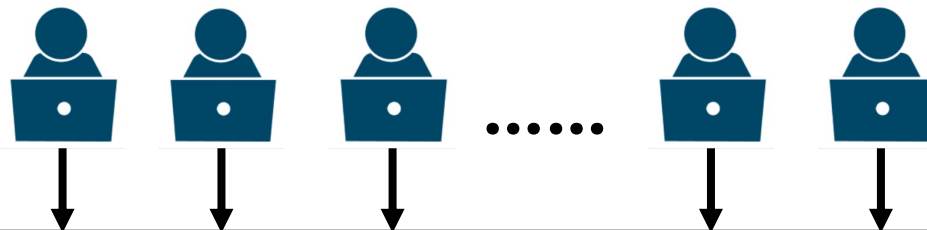


C clients

**Concurrent
Queries**



Dispatch each query to a worker thread

Per-query search flow ($\times T$ worker threads)

Start from the entry point

Beam-search
node expansion

Update candidate list
using PQ vectors in RAM

**Convergence-Aware
Early-Termination (§4.3)**

Top-k
converged

Re-rank with
full-precision vector

Return top-k results

Top-k not
converged

$\times \min(C, T)$

Shared across worker threads

**Popularity-Adaptive
Cache Manager (§4.2)**

Node already in cache

Node not
in cache

Shared NVMe SSD
(queued for retrieval)

Update cache (CLOCK)

